A standardized workflow for computer-assisted language comparison

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Language Science Meets Linguistic Diversity



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Introduction

Goals

Content

- Present you a workflow that is easy to use for your own study

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  1. Data digitization

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- Present you a workflow that is easy to use for your own study
  1. Data digitization
  2. The role of CLDF

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  3. Annotating comparative wordlists

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  2. The role of CLDF
  3. Annotating comparative wordlists
  4. Further steps

Cross-Linguistic Data Formats (CLDF)

CLDF



Why?

To allow exchange of cross-linguistic data and decouple development of tools and methods from that of databases, standardized data formats are necessary.

Once established, these dataformats could become a foundation not only for tools but also for instruction material in the spirit of Data Carpentry for historical linguistics and linguistic typology.

What?

The main types of cross-linguistic data we are concerned with here are any tabular data which is typically analysed using quantitative (automated) methods or made accessible using software tools like the `clld` framework, such as

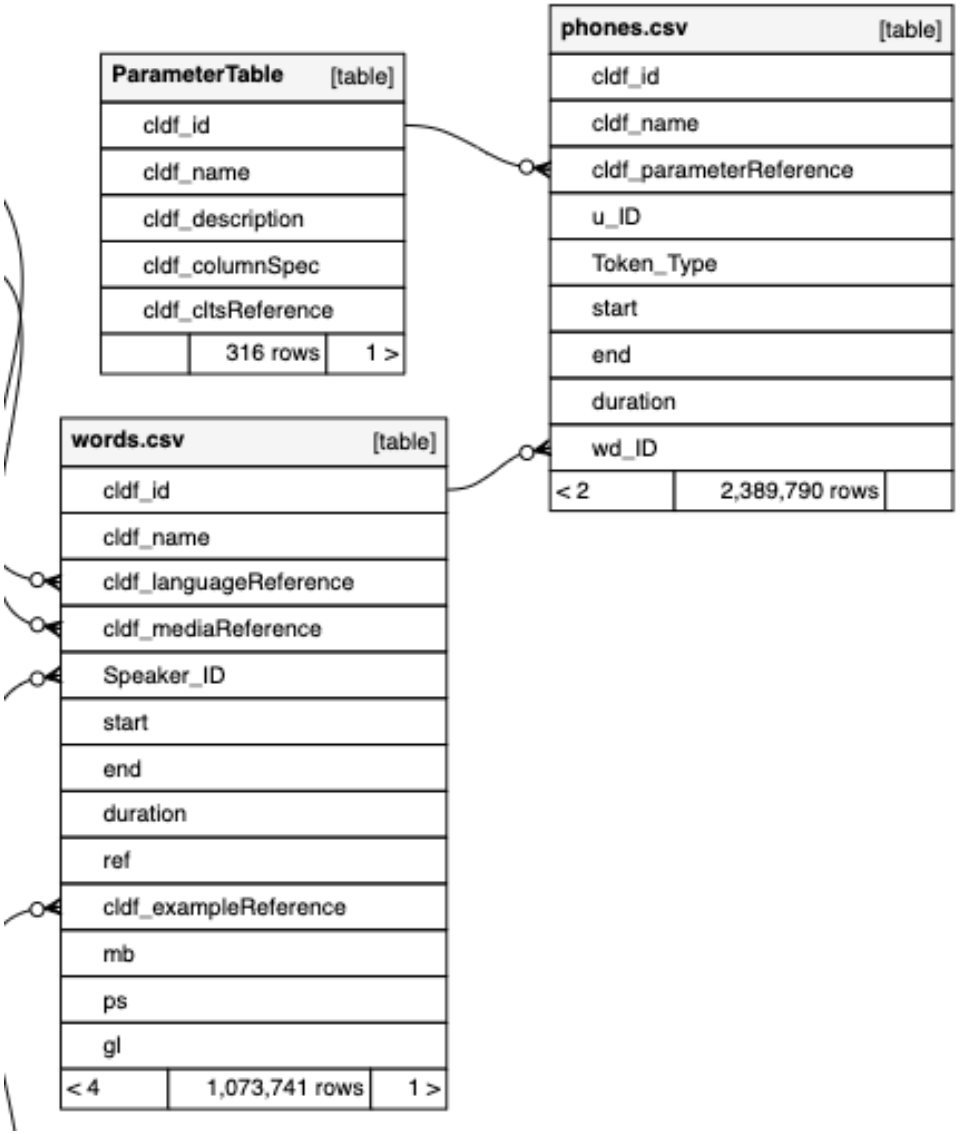
- wordlists (or more complex lexical data including e.g. cognate judgements),
- structure datasets (e.g. WALS features),
- simple dictionaries.

CLDF

Features

- One value per cell
- Multiple tables
- Standardization ensures comparability
- Validation through provided software

CLDF


ParameterTable [table]
cldf_id
cldf_name
cldf_description
cldf_columnSpec
cldf_cltsReference
316 rows
1 >
phones.csv [table]
cldf_id
cldf_name
cldf_parameterReference
u_ID
Token_Type
start
end
duration
wd_ID
< 2
2,389,790 rows
words.csv [table]
cldf_id
cldf_name
cldf_languageReference
cldf_mediaReference
Speaker_ID
start
end
duration
ref
cldf_exampleReference
mb
ps
gl
< 4
1,073,741 rows
1 >

CLDF

Reference catalogues

- Glottolog
- CLTS
- Concepticon

Glottolog

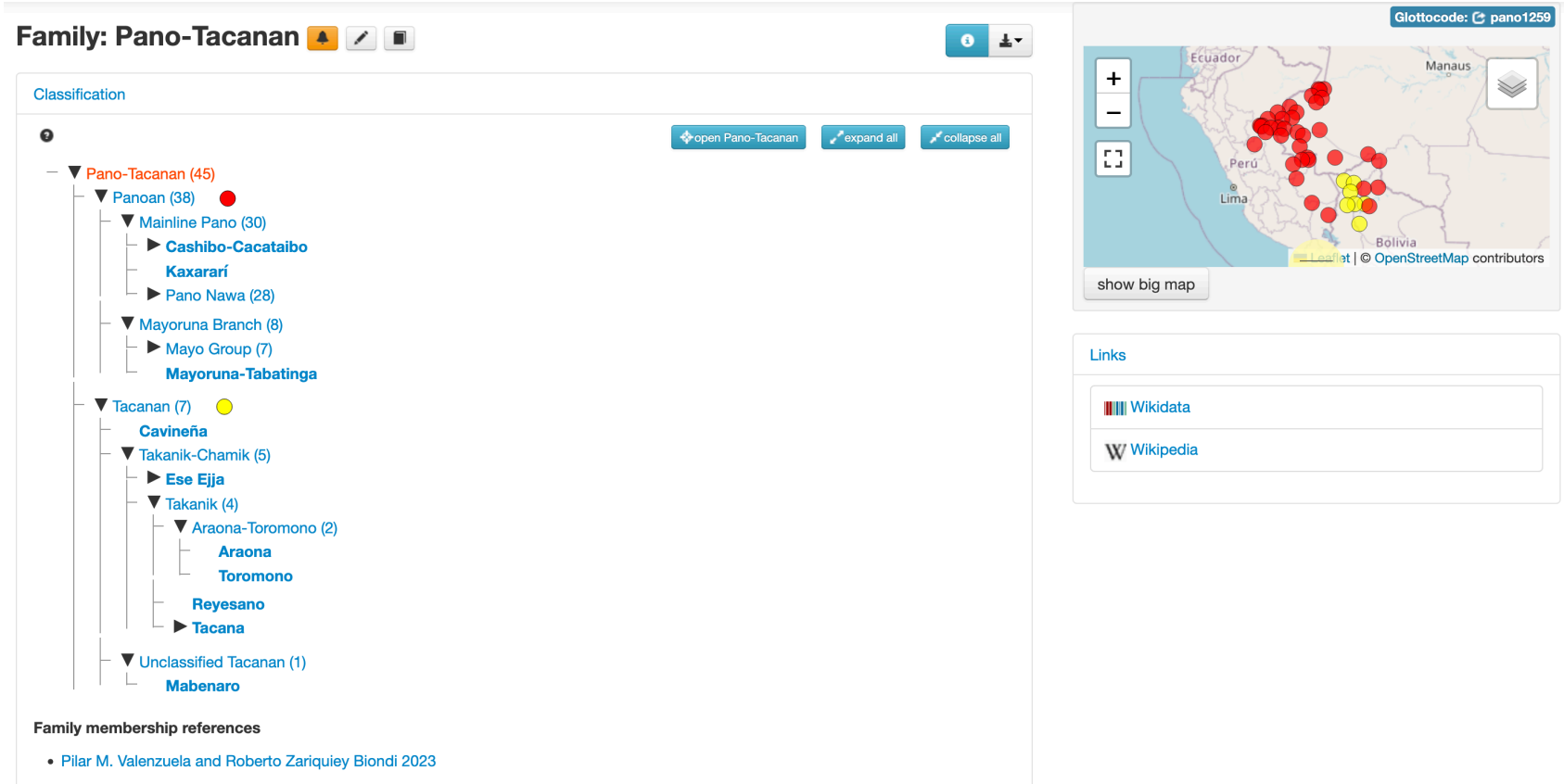
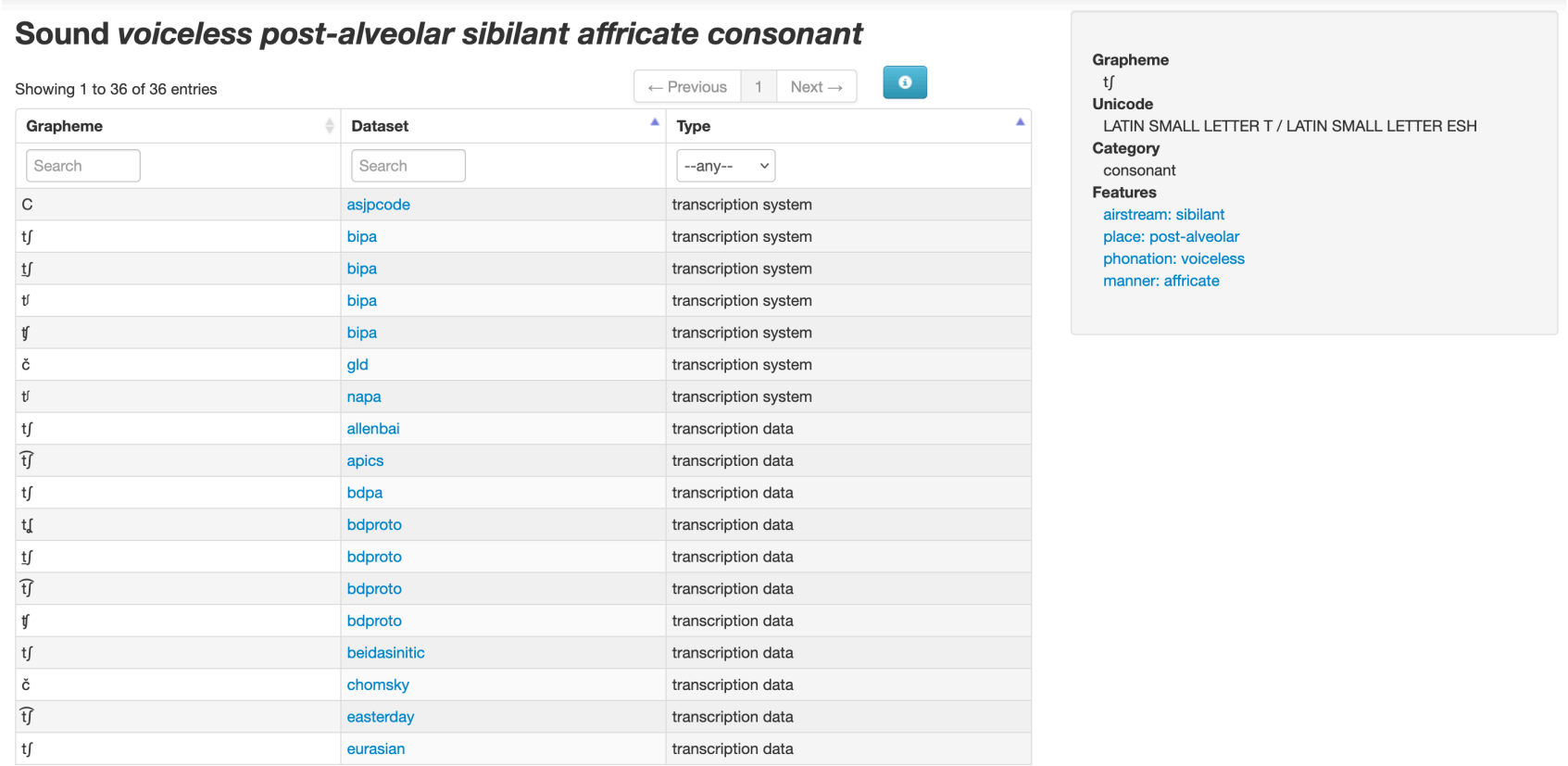


Figure 1: Metadata for languages (Hammarström et al. 2026)



Sound *voiceless post-alveolar sibilant affricate consonant*

Showing 1 to 36 of 36 entries ← Previous 1 Next →

Grapheme	Dataset	Type
Search	Search	--any--
C	asjpcode	transcription system
tʃ	bipa	transcription system
t͡ʃ	bipa	transcription system
tˈ	bipa	transcription system
ʧ	bipa	transcription system
č	gld	transcription system
tˈ	napa	transcription system
tʃ	allenbai	transcription data
t͡ʃ	apics	transcription data
tʃ	bdpa	transcription data
tʃ	bdproto	transcription data
tʃ	bdproto	transcription data
t͡ʃ	bdproto	transcription data
ʧ	bdproto	transcription data
tʃ	beidasinitic	transcription data
č	chomsky	transcription data
t͡ʃ	easterday	transcription data
tʃ	eurasian	transcription data

Grapheme
tʃ
Unicode
LATIN SMALL LETTER T / LATIN SMALL LETTER ESH
Category
consonant
Features
airstream: sibilant
place: post-alveolar
phonation: voiceless
manner: affricate

Figure 2: Metadata for sounds and phonetic features (List et al. 2024)

Concepticon

Concept lists

Showing 1 to 13 of 13 entries (filtered from 466 total entries)

← Previous | 1 | Next →

Note	Name	Compiler	Alias	Items	Tags	Uniqueness	Year	Gloss languages	Target languages	Sources
	swades		Search	Search	--any--	Search	Search	--any--	Search	
more	Swadesh 1952 200	Swadesh, Morris	Swadesh-200	200	basic	0.01	1952	english	Global	Swadesh 1952
more	Swadesh 1964 100	Swadesh, Morris	Swadesh-100-Russian	100	basic	0.00	1964	english russian	Global	Swadesh 1964
more	Swadesh 1955 100	Swadesh, Morris	Swadesh-100	100	basic	0.00	1955	english	Global	Swadesh 1955
more	Swadesh 1960 215	Swadesh, Morris		215	basic	0.01	1960	english russian	Global	Swadesh 1964
more	Swadesh 1954 97	Swadesh, Morris		97	basic	0.00	1954	english	American Indian languages	Swadesh 1954
more	Swadesh 1971 100	Swadesh, Morris		100	basic	0.00	1971	english	Global	Swadesh 1971
more	Swadesh 1960 100a	Swadesh, Morris		100	basic	0.00	1960	english	Global	Hymes 1960
more	Swadesh 1960 200	Swadesh, Morris		200	basic	0.01	1960	english	Global	Hymes 1960
more	Swadesh 1950 215	Swadesh, Morris		215	basic	0.01	1950	english	Global	Swadesh 1950
more	Swadesh 1955 215	Swadesh, Morris		215	ranked	0.01	1955	english	Global	Swadesh 1955
more	Swadesh 1960 100b	Swadesh, Morris		100	unstable	0.01	1960	english	Global	Hymes 1960
more	Swadesh 1950 165	Swadesh, Morris		165	areal	0.01	1950	english	Salish	Swadesh 1950

Figure 3: Metadata for conceptlists (List et al. 2025)

Concepticon

Showing 1 to 100 of 321 entries

← Previous | 1 | 2 | 3 | 4 | Next →

Id	Concept in source	Conceptlist
Search	Search	Search
Abraham-2018-307-87	fire [english]	Abraham 2018 307
Allen-2007-500-24	火 [chinese]; fire [english]	Allen 2007 500
Alpher-1999-151-29	fire [english]	Alpher 1999 151
Anonby-2018-1500-1112	fire [english]; آتش [persian]	Anonby 2018 1500
Araujo-1996-289-69	Feuer [german]; fogo [portuguese]	Araujo 1996 289
Ardila-2007-40-14	Su [basque]; Fire [english]; Fajro [esperanto]; Iso [guahibo]; Ignis [latin]; Ogon [russian]; Fuego [spanish]; Ates [turkish]	Ardila 2007 40
Auderset-2023-240-134	FIRE [english]	Auderset 2023 240
Backstrom-1992-210a-55	fire [english]	Backstrom 1992 210a
Backstrom-1992-210b-55	fire [english]	Backstrom 1992 210b
Backstrom-1992-210c-55	fire [english]	Backstrom 1992 210c
Backstrom-1992-210d-55	fire [english]	Backstrom 1992 210d
Backstrom-1992-210e-55	fire [english]	Backstrom 1992 210e
Backstrom-1992-210f-55	fire [english]	Backstrom 1992 210f
Bastin-1999-92-27	feu [french]	Bastin 1999 92
Bender-1971-99-27	fire [english]	Bender 1971 99
Benedict-1976-100-82	fire [english]	Benedict 1976 100

CLDF datasets: DoReCo

Corpus files : download languages metadata download citations in bibtex

Showing 1 to 53 of 53 entries ← Previous 1 Next →

Language	Glottocode	Family	Area	Creator(s)	License(s)	Word tokens	Speakers	Texts	Gloss
Anal	anal1239	Sino-Tibetan	Eurasia	Ozerov, Pavel	CC BY, CC BY-NC	13190	12	23	none
Arapaho	arap1274	Algic	North America	Cowell, Andrew	CC BY	4612	4	11	all
Asimjeeg Datooga	tsim1256	Nilotic	Africa	Griscom, Richard	CC BY	8782	12	40	none
Baïnounk Gubëeher	bain1259	Atlantic-Congo	Africa	Cobbinah, Alexander Yao	CC BY	11423	9	17	some
Beja	beja1238	Afro-Asiatic	Africa	Vanhove, Martine	CC BY-NC	15427	5	58	all
Bora	bora1263	Boran	South America	Seifart, Frank	CC BY	8364	6	9	all
Cabécar	cabe1245	Chibchan	North America	Quesada, Juan Diego and Skopeteas, Stavros and Pasamonik, Carolina and Brokmann, Carolin and Fischer, Florian	CC BY-NC, CC BY-NC-ND	10613	10	39	all
Cashinahua	cash1254	Pano-Tacanan	South America	Reiter, Sabine	CC BY	9655	3	7	some
Daakie	port1286	Austronesian	Papunesia	Krifka, Manfred	CC BY	11880	10	16	some
Dalabon	ngal1292	Gunwinyguan	Australia	Ponsonnet, Maïa	CC BY	3340	4	10	some
Dolgan	dolg1241	Turkic	Eurasia	Däbritz, Chris Lasse and Kudryakova, Nina and Stapert, Eugénie and Arkhipov, Alexandre	CC BY, CC BY-NC	8775	6	9	all
English (Southern England)	sout3282	Indo-European	Eurasia	Schiborr, Nils Norman	CC BY	8914	2	4	all
Evenki	even1259	Tungusic	Eurasia	Kazakevich, Olga and Klyachko, Elena	CC BY	8310	23	36	all
Fanbyak	orko1234	Austronesian	Papunesia	Franjieh, Michael	CC BY	10162	10	26	some

Figure 5: Corpus data in CLDF (Blum et al. 2024; Seifart et al. 2024).

LexiBank

A collection of datasets

Id	Name	# languages	# concepts	# words
Search	Search	Search	Search	Search
aaleykusunda	CLDF dataset derived from Aaley and Bodt's "New Kusunda data: A list of 250 concepts" from 2020	1	227	227
abrahammonpa	CLDF dataset derived from Abraham et al.'s "Sociolinguistic Research on Monpa" from 2018	12	304	3,208
allenbai	CLDF dataset derived from Allen's "Bai Dialect Survey" from 2007	9	499	4,546
backstromnorthernpakistan	CLDF dataset derived from Backstrom and Radloff's "Sociolinguistic Survey of Northern Pakistan" from 1992	5	219	993
bantubvd	CLDF dataset derived from Greenhill and Gray's "Bantu Basic Vocabulary Database" from 2015	10	415	4,187
beidasinitic	CLDF dataset derived from Beijing University's "Chinese Dialect Vocabularies" from 1964	18	736	14,745
birchallchapacuran	CLDF dataset derived from Birchall et al.'s "A Combined Comparative and Phylogenetic Analysis of the Chapacuran Language Family" from 2016	3	124	355
blustaustronesian	CLDF dataset derived from Blust's Austronesian data coded for the Austronesian Basic Vocabulary Database from 2008	401	210	83,873
bodtkhobwa	CLDF dataset derived from Bodt's "Lexical Cognates in Western Kho-Bwa" from 2019	8	542	4,264
bowernpny	CLDF dataset derived from Bowern and Atkinson's "Internal Structure of Pama-Nyungan" from 2012	147	338	36,513
bremerberta	CLDF dataset derived from Bremer's "Sociolinguistic Survey of Six Berta Speech Varieties in Ethiopia" from 2016	3	183	558

Figure 6: Individual datasets adhering to diffrerent criteria.

Publication

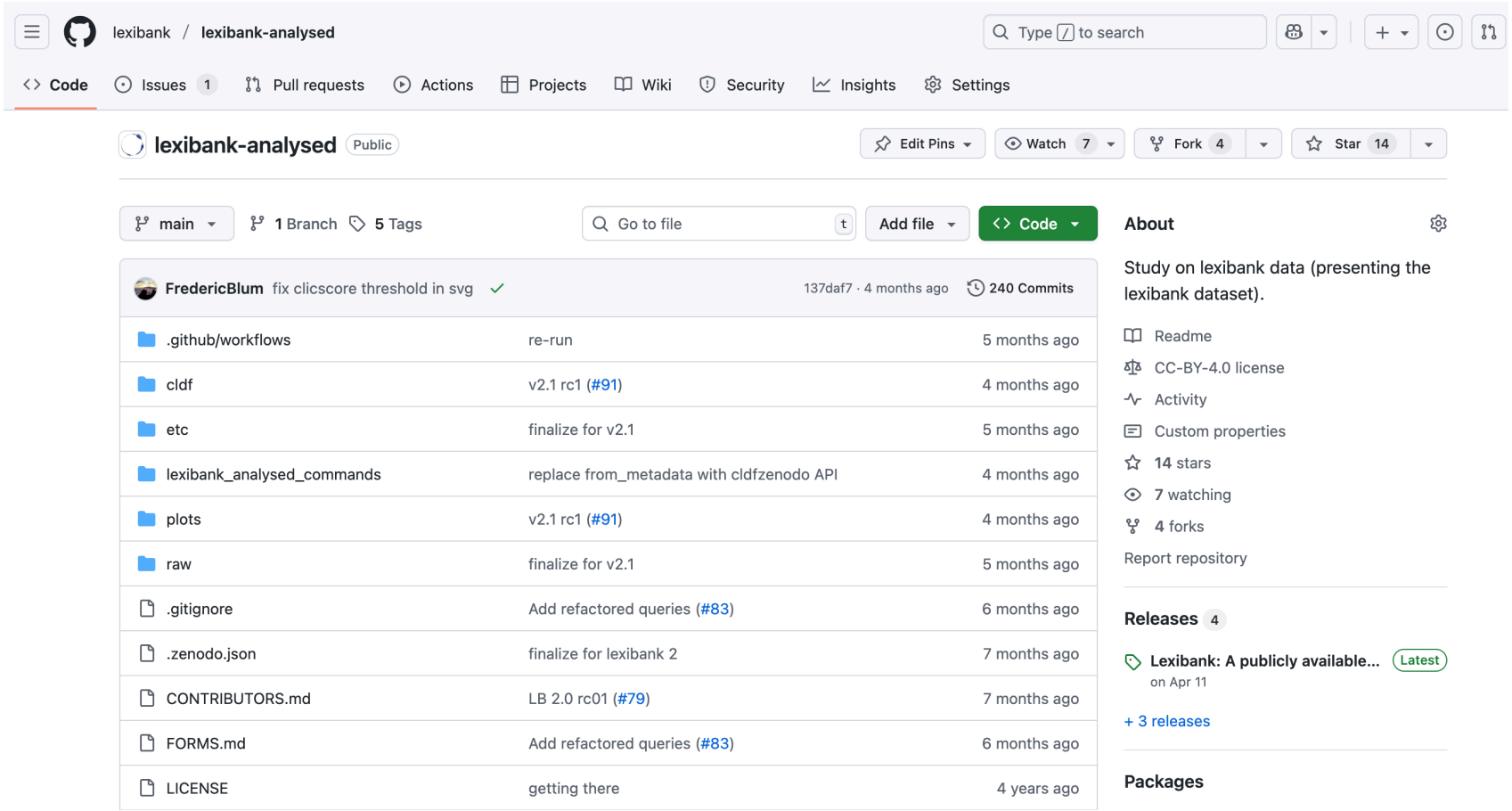


Figure 7: OpenAccess data.

LexiBank datasets

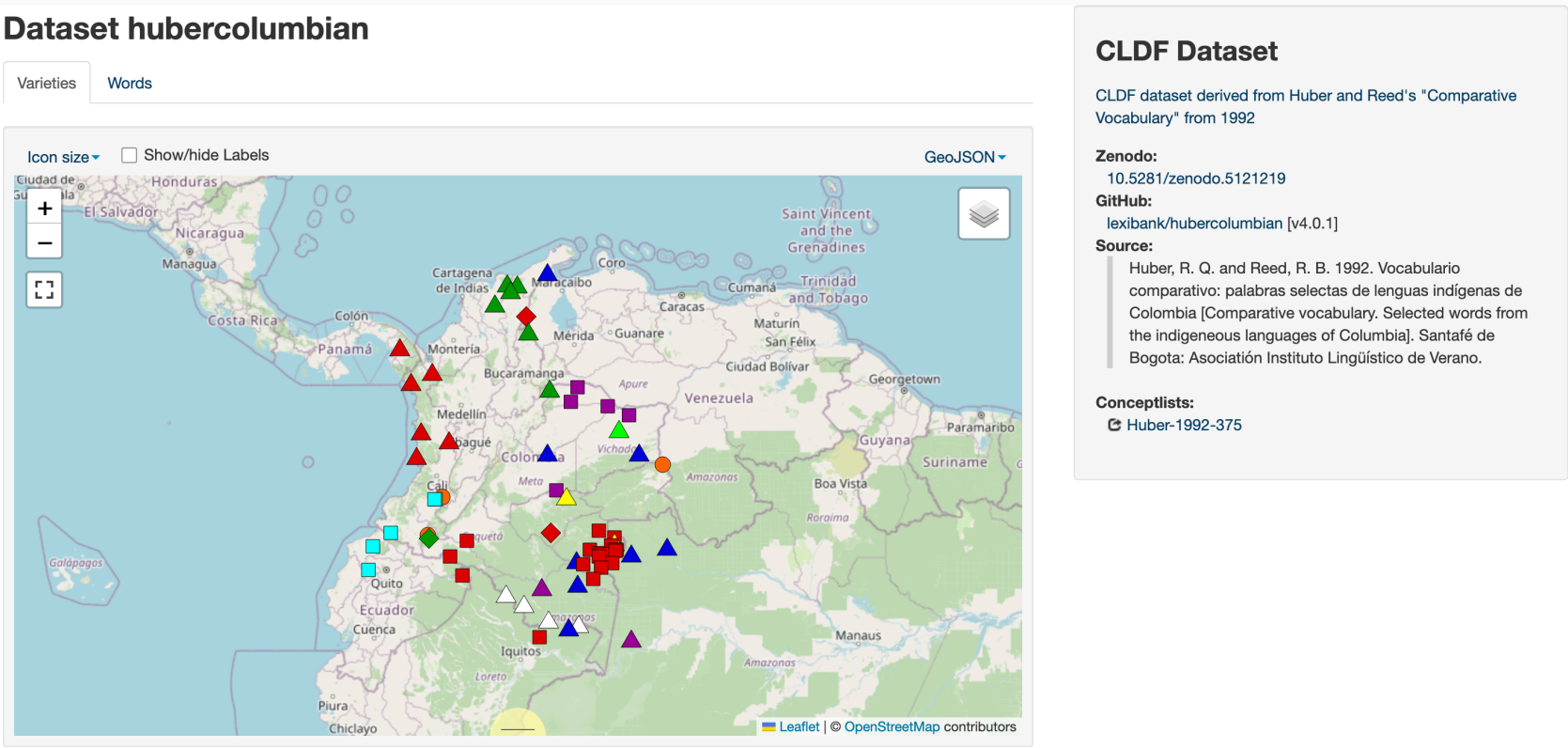


Figure 8: One example of a LexiBank dataset.

LexiBank data

Form	Concept	Segments	Variety	Family
Search	Search	Search	Search	--any--
a	HAIR (HEAD)	a	Chimila	Chibchan
a	HE	a	Ika	Chibchan
a	SHE	a	Ika	Chibchan
a	SEED	a	Orejón	Tucanoan
a	HAIR (HEAD)	a	Tsafiqui Pila	Barbacoan
a	HAIR (BODY)	a	Tsafiqui Pila	Barbacoan
a	IF	a	Wanano	Tucanoan
a¹kaji²	FISH	a k a j i	Nukak	Kakua-Nukak
aa	COME	aː	Ocaina	Huitotoan
aá	WALK	a a	Tatuyo	Tucanoan
áá	WALK	a a	Carapana	Tucanoan
áá	THEY	a a	Ocaina	Huitotoan
ãã	SAY	ã ã	Ocaina	Huitotoan
ãã	DOWN	ã ã	Ocaina	Huitotoan
ãá	NOT	ã ã	Ocaina	Huitotoan
aaa	RIPE	aː a	Ocaina	Huitotoan
áaáamo	SHE	a a a a m o	Muinane	Boran
aabare	LONG	aː b a r e	Cha'palaachi	Barbacoan
áabata	OTHER	a a b a t a	Achagua	Arawakan

Figure 9: One example of a LexiBank dataset.

Colexifications in LexiBank

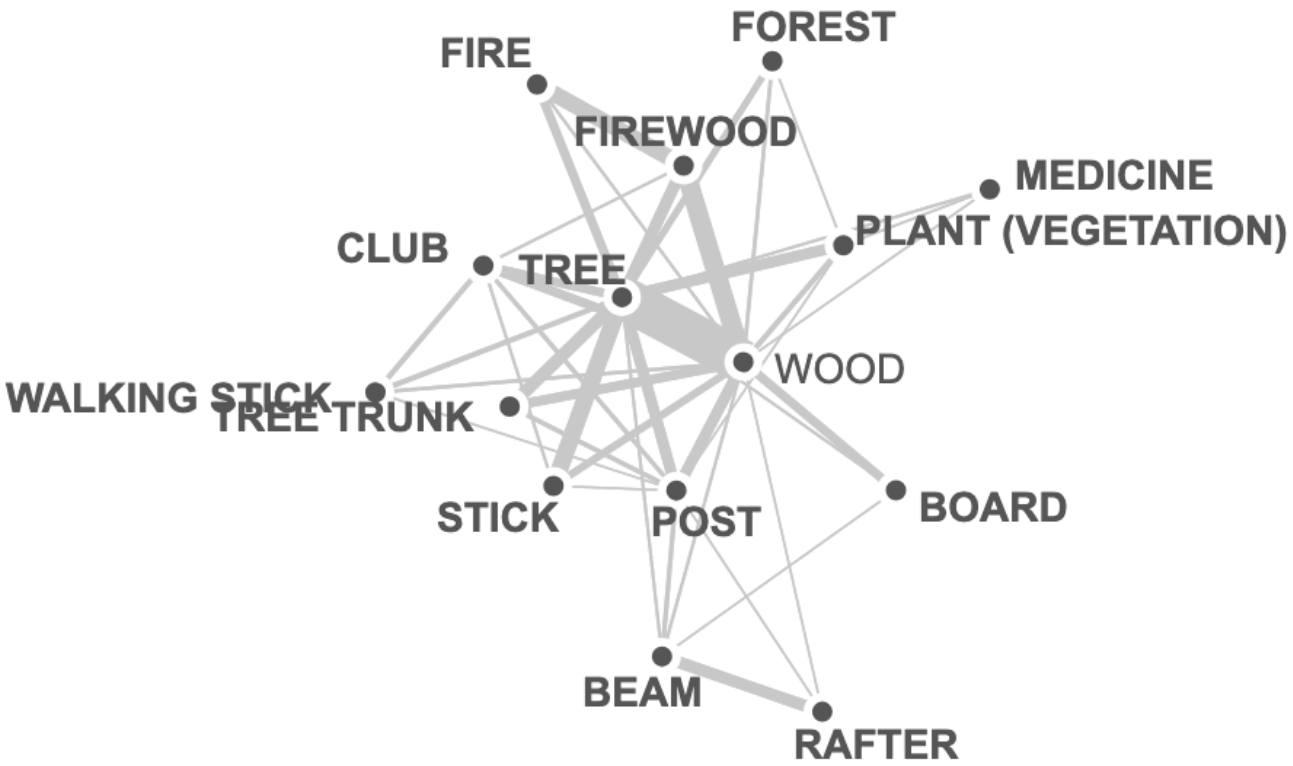


Figure 10: Colexification network inferred from LexiBank dataset.

Data digitization

5. *ʔano ‘paca (*Cuniculus paca*)’ : Ksh ʔano : SK ʔano : Kp ʔano : Mar -- : Ch ʔano ‘jochi pintado’ (ZINGG, 1998) : Kax anawɨ : Yam ano, anõ ‘majás’ : Chan -- : Shar ano, anon ‘majás’ : Shan anu : Kat ano : Poy -- : A ʔanó ‘majás, roedor pequeño’ (HYDE, 1980), aˈno I : Kn anu : M ano ‘paca, clase de roedor’ : Yaw anu : Ko -- : Mt -- : My --.

Figure 11: Reconstruction of Proto-Pano (Oliveira 2014).

Data digitization

DOCULECT	CONCEPT	ENTRY IN SOURCE	ALIGNMENT	PROTO SET
Proto-Panoan	paca	*?ano ‘paca (Cuniculus paca)’	?ano---	§5
Amawaka	paca	A ?anó ‘majás;; roedor pequeño’ (HYDE;;…	?an!óo---	§5
Amawaka	paca	A ?anó ‘majás;; roedor pequeño’ (HYDE;;…	!?ano---	§5
Chakobo	paca	Ch ?ano ‘jochi pintado’ (ZINGG;; 1998)	?ano---	§5
Katukina	paca	Kat ano	-ano---	§5
Kaxarari	paca	Kax anawɨ	-anawɨ-	§5
Kaxinawa	paca	Kn anu	-anu---	§5

Figure 12: Digitization of Oliveira (2014) (Blum & Barrientos 2023).

Conclusion

Why should I do all this?

Advantages

- Higher quality of data
- Accessible data that is readable for others
- Comparability of data ensures interoperability
- Large datasets for quantitive or comparative studies

. . . in practice

Outlook

References I

Blum, Frederic & Carlos Barrientos. 2023. A new dataset with phonological reconstructions in CLDF. *Computer-Assisted Language Comparison in Practice* 6(1). 43–51. https://doi.org/10.15475/calcip.2023.1.6.

Blum, Frederic et al. 2024. Consonant lengthening marks the beginning of words across a diverse sample of languages. *Nature Human Behaviour* 8. 2127–2138. https://doi.org/10.1038/s41562-024-01988-4.

Hammarström, Harald et al. 2026. *Glottolog database (v5.3)*. Leipzig: Max Planck Institute for Evolutionary Anthropology. https://doi.org/10.5281/zenodo.18840935.

List, Johann-Mattis et al. 2024. *CLTS. Cross-Linguistic Transcription Systems (v2.3.0)*. https://doi.org/10.5281/zenodo.10997741.

List, Johann-Mattis et al. 2025. *CLLD Concepticon (v3.4.0)*. Leipzig: Max Planck Institute for Evolutionary Anthropology. https://doi.org/10.5281/zenodo.14923561.

Oliveira, Sanderson Castro Soares de. 2014. *Contribuições para a reconstrução do Protopáno*. Brasília. https://repositorio.unb.br/handle/10482/17129.

Seifart, Frank, Ludger Paschen & Matthew Stave. 2024. *Language Documentation Reference Corpus (DoReCo)*. Version 2.0. Lyon. https://doi.org/10.34847/nkl.7cbfq779.